

CB-Signal: A Monetary-Policy Tone Signal from Central Bank Communications

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- Central bank tone — hawkish vs. dovish — is argued to carry information beyond the policy decision itself.
- Question: does a systematic, text-derived tone signal, built with openly documented methods and tested on **real market data**, produce a backtestable edge once transaction costs are applied?
- We answer this plainly, including when the answer is “not robustly.”

- **3,968 real documents**, 1997–2026, scraped directly from issuing institutions:
 - ECB: 1,871 speeches
 - Fed: 1,649 (statements, minutes, speeches)
 - BoE: 448 (speeches, MPC minutes)
- **Rates**: 2Y/10Y US Treasury, DE/UK long yields, VIX — FRED, 1956–2026 (39,665 obs)
- All three central-bank scrapers required real fixes this cycle: ECB/BoE listing pages moved to JS-rendered search widgets; Fed's meeting-date discovery had a pre-existing bug mislabelling minutes-release dates as meeting dates.

Three independent sentiment tiers

- 1 **Dictionary**: curated hawkish/dovish lexicon (Apel & Blix Grimaldi 2012; Hansen & McMahon 2016; Bennani & Neuenkirch 2017) + negation handling.
- 2 **FinBERT** (ProsusAI): pretrained financial-sentiment transformer, chunked & averaged over long documents.
- 3 **Fine-tuned tier**: MiniLM sentence embeddings + logistic regression, trained on a small *author-curated* (not human-annotated) 482-example set.

Cross-checking three independently-built methods makes the dictionary baseline's results harder to dismiss as a lexicon artifact.

Signal construction and backtest

- Weekly hawkish index $\rightarrow$ 12-week EMA baseline $\rightarrow$ expanding-window z-scored surprise (point-in-time by construction).
- Tradable signal: 4-week rolling blend of level & surprise, applied to 2Y/10Y bond-return proxies.
- 20bps round-trip transaction costs, $T+1$ execution.
- **Found & fixed:** the original regime-labelling HMM was fit once on the full sample (look-ahead for that label). A walk-forward refit changes 45% of weekly labels. It doesn't affect the backtest P&L — regime is descriptive only — but it did affect how trustworthy the regime plot was.

Full-sample results: no robust edge

Strategy	Ann. return	Sharpe	Max DD	Turnover/yr
Short 10Y	+0.07%	0.07	−6.1%	2.2
50/50	−0.24%	−0.40	−10.0%	2.9
Short 2Y	−0.55%	−1.61	−15.2%	3.6

Weaker than the synthetic-data version of this same pipeline (Sharpe up to 1.75) — expected, since synthetic tone is constructed to drive returns.

But: the signal works in one specific regime

Period	Strategy	Sharpe
GFC (2007–09)	50/50	−1.35
ZIRP (2009–15)	50/50	−1.18
COVID (2020–21)	50/50	−1.01
Hiking (2022–23)	50/50	1.83
Plateau (2024+)	50/50	−1.12

Hypothesis: tone carries more information when policy stance is actually *changing* quickly, less when anchored (ZIRP) or shock-dominated (COVID). One episode — suggestive, not proven.

Limitations

- Negative-to-flat full-sample result — reported because it's true.
- Fine-tuned tier's training set is small and author-curated, not independently annotated.
- Sub-period finding rests on a single hiking cycle.
- Constant-duration bond-return proxy, not live futures/contract specs (2Y leg likely uneconomic at 20bps TC on cash bonds; futures round-trips are closer to 1bp).

Reproduce

- Code, data, paper, and this deck:
`github.com/SakaiMarouane/cb-signal`
- Full real-data run (scrape + score + backtest), < 2 hours:

```
python -m cb_signal.ingest
python -m cb_signal.run_all --skip-synth
        --subperiods --walk-forward
```